

Airbnb Price Prediction Report - Milan

Generated on 2026-08-25 by the automated pipeline (*make all*).

Executive summary

The pipeline trained and compared Random Forest, XGBoost and TabPFN on Airbnb listing data for **Milan** (14317 listings). The **xgboost** model was selected for final evaluation on a held-out test set and achieved: - **RMSE: \$107** - **MAE: \$54** - **R²: 0.50**

Data

- Source: InsideAirbnb (latest available snapshot). See: <https://insideairbnb.com/get-the-data/>
- City: Milan · 14317 listings after cleaning

EDA

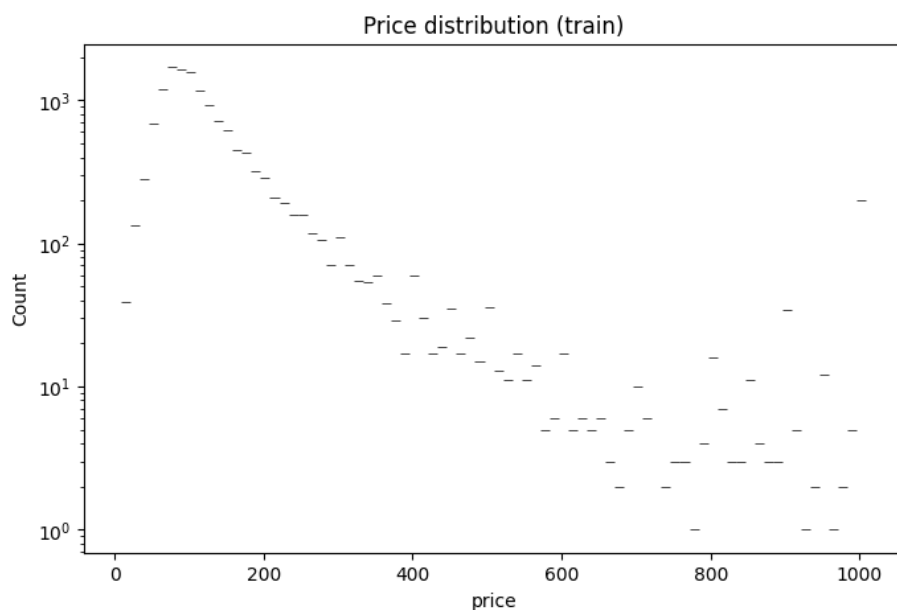
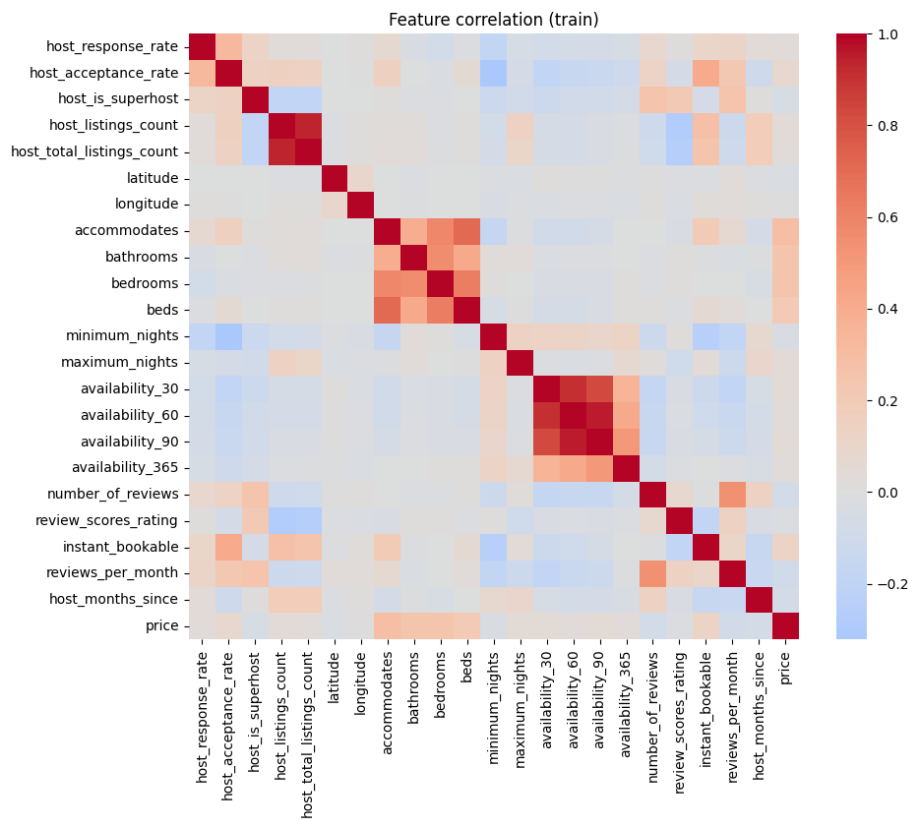
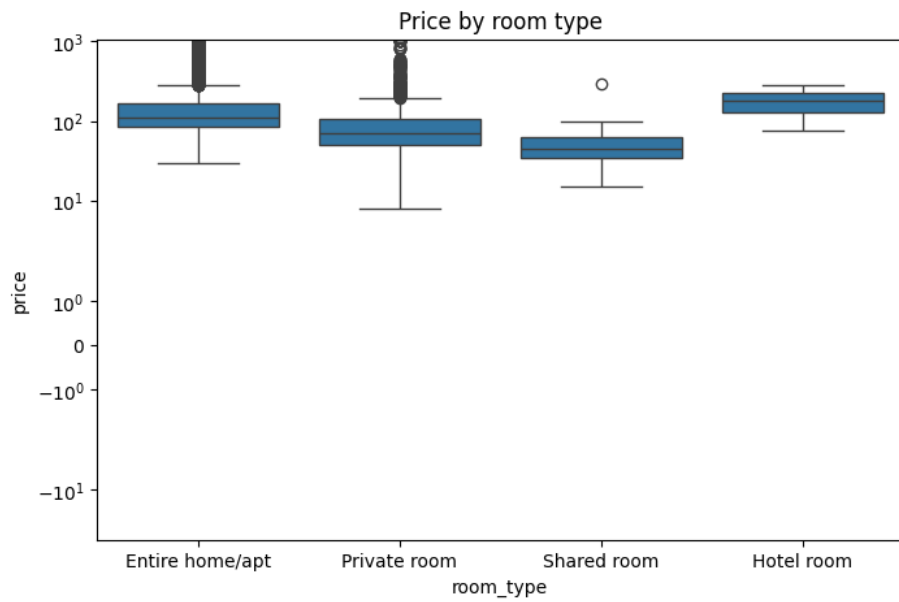


Figure 1: price_dist

- Median price: **\$108** (mean \$152)



Models & Results

Model	Validation RMSE
Random Forest	\$121
XGBoost	\$112
TabPFN (capped sample, not directly comparable)	\$108

Holdout results (best model): see `artifacts/milan/holdout_results.csv`.

Top features

Feature	Importance
room_type_Entire home/apt	0.053
room_type_Private room	0.052
bedrooms	0.045
bathrooms	0.044
instant_bookable	0.030
host_response_time_nan	0.030
accommodates	0.026
host_neighbourhood_South Kensington	0.026
host_listings_count	0.025
host_total_listings_count	0.025

Recommendations

- The best-performing model was **xgboost** (holdout RMSE \$107, MAE \$54, R^2 0.50).
- On validation, **XGBoost** beat Random Forest by 7% on RMSE (\$112 vs \$121). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **room_type_Entire home/apt, room_type_Private room, bedrooms** — i.e. room_type_Entire home/apt (the room type (entire home vs private/shared room) (specifically Entire home/apt)); room_type_Private room (the room type (entire home vs private/shared room) (specifically Private room)); bedrooms (the number of bedrooms). Hosts can use these levers when setting nightly rates.
- Accuracy caveat: the holdout RMSE of \$107 is **99% of the median listing price (\$108)**, so predictions for typical listings are expected to be off by roughly $\pm$ \$107.

- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

*Automated pipeline: fetch → preprocess → EDA → train → evaluate → report.
Code in **src/**, config in **configs/base.yaml**.*